

0c_Scaling_Data

April 10, 2026

1 Test Scaling Data

```
[ ]: import sys
      sys.path.insert(0, '..')

      import numpy as np
      import pandas as pd
      import matplotlib.pyplot as plt

      from utils.data import load_and_clean_data, get_train_test_split,
      ↪ Feature_Min_Max_Scaler, Min_Max_Scaler, Standard_Scaler
      from utils.evaluation import evaluate_model, add_result
      from utils.plotting import plot_features_vs_target, plot_predicted_vs_actual,
      ↪ plot_residuals, plot_feature_importances

      plt.rcParams['figure.dpi'] = 100
      %matplotlib inline
```

1.1 MinMax_Scaler von [0,1]

```
[ ]: df = load_and_clean_data()
      df_scaled_MM_0_1 = Min_Max_Scaler(df, min_val=0, max_val=1)
      fig, axes = plot_features_vs_target(df_scaled_MM_0_1,
      ↪ save_name="scaling_0_1_features_vs_target")
      plt.show()
```

1.2 MinMax_Scaler von [-1, 1]

```
[ ]: df_scaled_MM_neg1_1 = Min_Max_Scaler(df, min_val=-1, max_val=1)
      fig, axes = plot_features_vs_target(df_scaled_MM_neg1_1,
      ↪ save_name="scaling_neg1_1_features_vs_target")
      plt.show()
```

1.3 StandardScaler

```
[ ]: df_scaled_standard = StandardScaler(df)
fig, axes = plot_features_vs_target(df_scaled_standard,
    ↪save_name="scaling_standard_features_vs_target")
plt.show()
```

1.4 Feature scaler min max

```
[ ]: df_feature_scaled = Feature_Min_Max_Scaler(df, feature="MedInc", min_val=0,
    ↪max_val=1)
fig, axes =
    ↪plot_features_vs_target(df_feature_scaled,save_name="scaling_0_1_MedInc_features_vs_target")
plt.show()
```

```
[ ]:
```